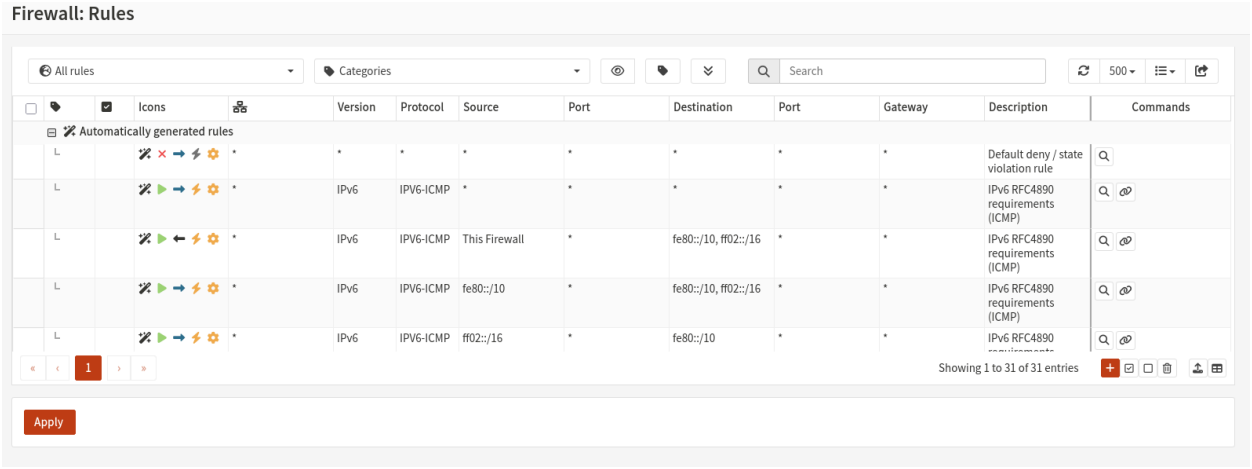
# Home Lab Project Week 9, Day 2

## Task:

Configure restrictive inbound firewall rules and establish a default-deny approach for the appropriate lab network segment.

### Default-deny on OPNsense (lab segment)

1. **Firewall → Rules → [your lab VLAN/segment interface]**
    - OPNsense, like pfSense, has an **implicit deny-all** at the bottom of every interface's rule set — you don't create that rule, it's baked in. the job is to make sure nothing above it is accidentally wide open.

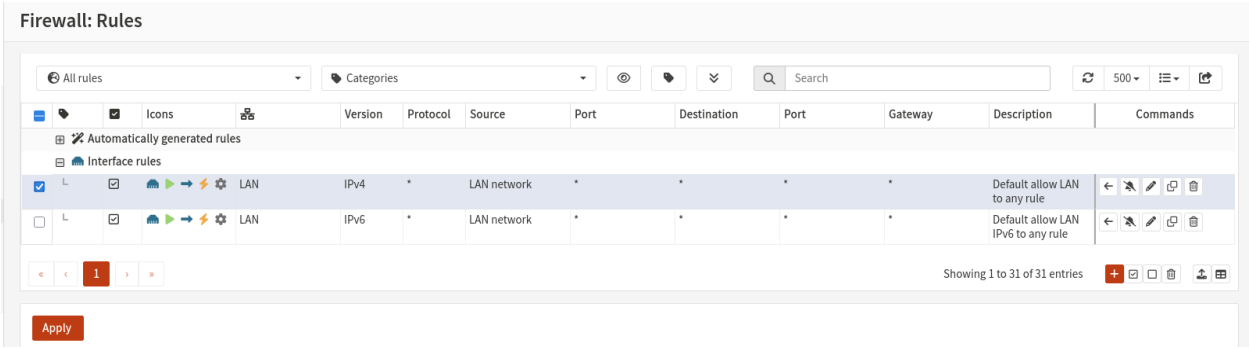




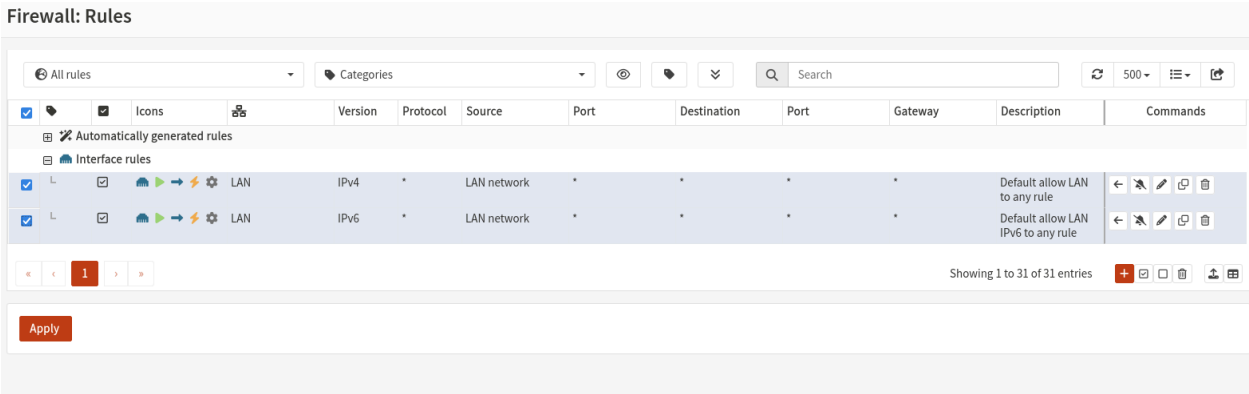


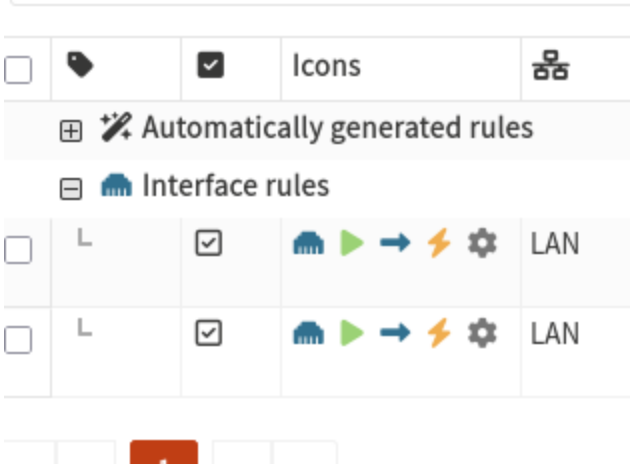
**Now remove the allow-any rule**

2. Collapse **Automatically generated rules** and expand **Interface rules**.
3. Disable the **IPv4** “Default allow LAN to any rule” by clicking the checkbox icon in the ☑ column. The row turns grey.




4. Do the same for the **IPv6** row.






5. Click **Apply**.
6. Refresh the GUI page. It should still load because of the anti-lockout rule.

7. **Check for (and remove) the default "allow LAN to any" rule**
    - Fresh OPNsense LAN-type interfaces often ship with a broad allow rule. Delete or disable it for this segment if you want true default-deny.
8. **Add explicit allow rules, top to bottom, most specific first:**
    - Example: allow lab segment → management VLAN on SSH only, from one source
        - **Action:** Pass
        - **Interface:** LAB_VLAN
        - **Protocol:** TCP
        - **Source:** 10.10.10.0/24 (or your specific host)
        - **Destination:** management subnet
        - **Destination port:** 22
    - Add one rule per service you actually need (DNS, NTP, etc.) — nothing else.
9. **Confirm the implicit deny catches the rest**
    - You can watch this happen live in **Firewall → Log Files → Live View** — anything not matched by your explicit rules shows up as blocked.
10. **Logging (recommended for your portfolio/activity doc)**
    - On each explicit rule, there's a "Log packets matched by this rule" checkbox — turn it on for a couple of key rules so you can screenshot evidence of allowed vs. denied traffic for your challenge writeup.

## One OPNsense-specific gotcha worth flagging

Rule order in OPNsense evaluates **top-down, first match wins** — same as pfSense — so double check your explicit allows sit above anything broader, and that there's nothing accidentally re-opening things below them (OPNsense sometimes auto-generates rules for

things like DHCP/DNS resolver on LAN-type interfaces — worth scrolling through the full list once to confirm nothing else is quietly passing traffic).